

Work orders: executing the CZB roadmap with a smaller model

2026-08-27. Jonas asked whether the roadmap's execution can move to a less expensive model, and what would facilitate that. This document is the facilitation: task cards precise enough that a capable but cheaper session can execute them without design judgment, with acceptance criteria that make success checkable and stop conditions that route genuine judgment calls back to Jonas or a stronger review session. The recommendation on when to use which model is in section 1; the cards follow.

1 Recommendation on the model split

My honest view: most of the remaining work is well-specified implementation against existing patterns, and a cheaper model executes that reliably *if* the specification carries the judgment. The split I would use:

- **Cheaper model, per these cards:** A.1, A.2, A.4, B.1–B.3 constructions, C, E.1, document rebuilds, and all reruns.
- **Stronger model (or Jonas):** interpreting A.2's bias-model comparison, A.3's pass/fail verdict, B.4's transfer readout, anything where a card's stop condition fires, and a periodic review pass over completed cards (the pattern that caught the paired-difference overstatement is worth keeping: execute cheap, review strong).
- Do not hand a cheaper session an open question ("does this look right?") — hand it a card. When a card's acceptance criterion cannot be met, the correct output is a short failure report, not an improvisation.

2 Standing rules for every card (read first, they are load conditions)

- 1 **Every number that will be quoted anywhere must come from a committed script** with its output file tracked. No in-session-only analyses — this rule exists because an uncommitted analysis produced an overstated headline once already (results doc §4.3b-ii, corrected 2026-08-27).
- 2 **Run** `python tests/test_surprise.py, test_comfortzone.py, test_causation.py, test_cutin.py` **before and after each card**; all must pass. New behavior gets new property tests in the same `check()` style.
- 3 **Never modify** `src/aidriver/preferences.py` **defaults** — released behavior is frozen; new behavior goes behind flags defaulting off.
- 4 **House style** for anything written for Jonas: US English, markdown as source, Word + PDF generated (`pandoc + docs/build_pdf.py`), no period ending a heading, opinions hedged as opinions. OneDrive locks: if a build fails with `PermissionError`, write a -vN name and say so.
- 5 **Environment:** Python 3.14, torch CPU-only, no LaTeX/GPU/gh. Long jobs checkpointed and restartable; background runners never piped through `tail`.
- 6 **Provenance discipline:** cite only verified references; mark anything unverified.

3 Task cards

Card S.1 — draft the `perform-research` skill (first task for the next session)

- **Goal:** a reusable, iteratively updatable skill that captures how research is performed in this collaboration — so the working discipline survives model changes, session changes, and time. Jonas asked for it explicitly, drafting delegated to the next (cheaper) session, with the instruction to also look at what exists elsewhere online and borrow with attribution.
- **Build:** draft `C:\Users\bargman\claude\skills\perform-research\SKILL.md` with frontmatter (name, description triggering on research execution, analysis tasks, fitting runs, and model handovers in this project family). Required contents, from this project's accumulated practice:
- **Reproducibility discipline:** every quoted number from a committed script with tracked output; regeneration commands logged; generated artifacts never hand-edited; the 2026-08-27 paired-difference correction as the cautionary tale.
- **Verification discipline:** property tests before and after every task; verify by two independent routes where feasible; results that contradict documented findings get surfaced, never silently reconciled.
- **Documentation discipline:** markdown as source of truth, Word/PDF generated; dated handover files per arc plus a standing handover `.md` entry point; decisions recorded with their reasoning and their location; the `jonas-academic-writing` conventions for anything addressed to Jonas.

- **Between-model handover protocol:** the entry sequence (handover chain → roadmap → work orders), one card per session, the escalation triggers and review gates (copy from `handover_2026-08-27.md` §8 — that list is the current authority), worklog appends, and the rule that a cheaper session executes cards and never improvises around a failed acceptance criterion.
- **Statistical practice:** uncertainty conventions stated and resampled completely (all variance sources); comparisons as differences; held-out validation for any model comparison; pre-stated decision rules for verdict experiments.
- **Research step:** search online for existing practice worth borrowing — multi- agent / multi-model handoff patterns, LLM-assisted research workflow guides, lab-notebook and reproducibility standards (e.g. from open-science communities), and any published skill/prompt libraries for research execution. Cite what is borrowed; mark unverified claims per house rules.
- **Accept:** the draft `SKILL.md` exists, is under ~300 lines, cites its sources, and ends with an explicit "for Jonas's review" note. **Do not treat the skill as active until Jonas approves it;** iterate on his comments in later sessions.
- **Stop if:** unsure whether some practice is project law or one session's habit — list the uncertain items in the draft for Jonas to rule on rather than guessing.

Card R.1 — review gate after A.2/A.3/A.4 (for the stronger model)

- **Goal:** close the queries the executing session raised, in severity order, and decide the two things it deliberately did not decide.
- **Read first:** `replication/czb/out/query_register.md` (generated — the work log is the source of truth), then `stage1_summary.md`, `stage2_summary.md`, `button_validation_summary.md`, then the three worklog entries dated 2026-08-27.
- **The two decisions that block downstream work:**
- **A.2.Q2 — the bias variant.** Group-level versus hierarchical lapse moves the between-driver spread $0.341 \rightarrow 0.209$, a 39% change in the quantity a percentile is made of, while held-out likelihood separates them by 0.6 units over 3 096 trials. The fitted per-driver lapse spread is large (sd 2.76, logit scale), so the hierarchical lapse absorbs variation the group model gave to the threshold — the opposite direction to the leakage argument in `docs/czb_validation_roadmap.md` §5.1, which should be revisited or corrected in the light of it. Deciding this fixes the headline percentile.
- **A.3.Q1 — the accumulator's specification.** The pre-registered verdict is FAIL, but the in-sample fit is worse than the simpler model it extends, and the cause is identified: noise accumulating through 15.1 s of empty pre-onset clip. Choose the remedy (leaky accumulator; accumulation starting where evidence exists; or something else), then A.3 and A.4 are re-run and their verdicts mean something. Until then neither FAIL should be quoted as evidence about the framework.
- **Also verify, not just accept:** that the numbers quoted in the three summaries come from the committed scripts (the failure mode with precedent here is a number that exists only in a transcript); and that A.2's percentile table is read with its two minor caveats attached (A.2.Q4, A.2.Q5).
- **Owed work the executing session named:** A.2.Q3 asks for the pre-onset predictive under *both* bias variants; only the better-fitting one was checked.
- **Accept:** every open query either resolved with a `RESOLVED <card>.Q<n>: . . .` line appended to the work log, or explicitly carried forward with a reason; the register regenerated; and if a decision changes a documented plan, that document corrected in place with a dated note.

Card A.1 — synthetic-recovery harness

- **Goal:** prove the stage-1 estimator recovers known parameters before touching real data.
- **Build:** `replication/czb/fit_recovery.py`. Simulate the Random cut-in design (43 participants × 18 cells × 4 repetitions) from the generative model in `docs/czb_fitting_plan.md` §2 stage 1: $c_i \sim \text{LogNormal}(\mu, \sigma)$, lapse floor b (both variants), $P = b + (1-b) \cdot \Phi((x - c_i)/\sigma_{\text{resp}})$ with x = the cell's `deficit_max` from `comfortzone.czb_data.random_cutin_trials()`. Fit by MAP in torch (L-BFGS) with Laplace SEs; also fit the `a_allowed` parameterization (threshold on `a_req_max`).
- **Accept:** over 20 simulated datasets, population μ and σ recovered within 2 Laplace SEs in ≥ 18 ; per-driver c_i shrinkage visible (posterior means closer to truth than raw per-driver MLEs in RMSE). Runtime under 10 minutes total.

- **Pitfalls:** identifiability of $(c, \sigma_{\text{resp}})$ scale — fix σ_{resp} 's prior scale from the pilot ($\approx 2\,200$ deficit units); lapse and threshold trade off at C1 — check the recovery specifically at b.
- **Stop if:** recovery fails after two honest attempts — report the failure mode, do not loosen the acceptance criterion.

Card A.2 — stage-1 hierarchical fit, both bias variants

- **Goal:** the population distribution of the boundary level, and the bias-model decision.
- **Build:** `replication/czb/fit_stage1.py`, reusing A.1's model code on the real trials. Fit all four combinations: $\{\text{deficit axis, a_req axis}\} \times \{\text{group-level b, hierarchical b_i}\}$. The ordered braking-expectation response enters as two nested thresholds (fitting plan §2); both levels free per roadmap §0b.
- **Report** (`replication/czb/out/stage1_summary.md`): population μ, σ with SEs; percentile table (50–95 in steps of 5) of the population distribution with CIs; held-out log-likelihood (leave-one-participant-out, 43 folds); posterior predictive $P(\text{intervene})$ per C1 cell against observed; fitted comfort and dread levels translated to `a_allowed` units and to steady-following THW at 20 m/s via `comfortzone.field.critical_thw`.
- **Accept:** all four fits converge; the summary table renders; LOPO likelihood distinguishes the bias variants or states that it cannot.
- **Stop if:** the C1 cells misfit under both bias variants (systematic sign pattern across criticalities) — that is the anticipation-leak question and needs review.

Card A.3 — the accumulator layer (review gate on its verdict)

- **Goal:** the pre-registered test: does time-integration close the within-scenario gap?
- **Build:** `replication/czb/fit_stage2.py`. Evidence = the per-frame deficit series (not the running max) from `stimulus_field`; linear accumulator with gain and noise, threshold from A.2's `c_i`; $P(\text{crossed by clip end})$ per trial via the closed-form Gaussian first-passage approximation or 200-path simulation per cell (cells share stimuli, so simulate per cell, not per trial). Motor latency fixed at 0.25 s; sensitivity refits at 0.15 and 0.35 s.
- **Accept / decision rule (pre-stated):** held-out (leave-one-criticality-out) RMSE on the 18-cell surface ≤ 0.11 counts as closing the gap (the 2D state rule's 0.120 and the design ceiling's ~ 0.10 bracket the target); > 0.13 counts as failure and is reported as evidence against the framework per the assessment.
- **Stop if:** the fitted gain wants to be so large the accumulator degenerates to the static threshold — report, since that itself answers the question.

Card A.4 — button-side validation

- **Goal:** predict the press-time densities from the Random-fitted model with one paradigm shift.
- **Build:** `replication/czb/validate_button.py` on `czb_data.button_cutin_trials()`. Fit δ once as a threshold shift and once as a gain shift; compare predicted vs observed press-time distributions per criticality (QQ plots to figures/, one page) and report which shape fits.
- **Accept:** the documented cross-paradigm excess (+0.26 at TTC6) is reproduced by the shifted model within 0.05 at the fixed-clip checkpoints.

Cards B.1–B.3 — transfer-scenario field constructions

Template for each: a loader in `src/comfortzone/<scenario>.py` following `cutin.py`'s structure (role assignment appropriate to the scenario — **by instructed vehicle, not lateral span**), a field check script in `replication/czb/` producing a per-criticality deficit figure, property tests, and the PS-versus-field correlation (the `ps_vs_field.py` pattern) as the acceptance readout.

- **B.1 cyclist overtake** (first): ego closes on a cyclist in lane; rear-end field nearly unchanged; criticality = lateral clearance 0.5/1/1.5 m, so the lateral term matters — use the actual clearance in `dy`, widths from the trace (cyclist ≈ 0.6 m). Accept: deficit ordered by clearance at C2–C5; cell-level PS Spearman ρ reported.
- **B.2 truck overtake / truck lateral:** ego passes a truck with clearance variants; the conflict is lateral, so `P_lane` runs on the *lateral closing* induced by the truck's drift (`truck_lateral_movement` traces) and the clearance term carries the statics. Accept: same criteria; also record whether the zero-deficit issue appears.
- **B.3 LTAP:** crossing geometry. Build arrival-time separation at the conflict zone (the intersection scenario's construction): compute each vehicle's time to the crossing-region entry/exit from the traces, and gate the collision/safety terms by predicted co-occupancy (the `P_lane` idea with lateral overlap replaced by zone co-occupancy). This card has real design content — **write the construction note first (1 page, the lane-entry-note pattern) and get it reviewed before coding.**

- **Stop for all three if:** a trace family's geometry does not fit the template (e.g. missing vehicle, broken dimensions beyond the documented truck case).

Card B.4 — the transfer test (review gate; do not run before A.3 and B.1–B.3)

- Fit on cut-in only (A.2/A.3 configuration frozen); predict each transfer scenario's response surface with no refit. Primary: no per-scenario shift. Secondary: one instruction shift per scenario. Comparators on identical folds: per-scenario 2D state rules (1/TTC-analogue + the scenario's own second observable), and — once specified with Jonas — the elliptical joint on the same observables.
- Output: one table, per scenario × model: held-out corr and RMSE against (a) chance and (b) the per-scenario-refit ceiling. No verdict prose from the executing session; the verdict is a review task.

Card C — percentile sensitivity

- `replication/czb/percentile_sensitivity.py` from A.2's population distribution: implied trigger onset per stimulus for percentiles 50–95 (step 5) × `a_allowed` on the fitted comfort–dread range; report `d(onset)/d(5 percentile points)` in seconds against the percentile's own CI width. One figure, one table.

Card E.1 — closed-form versus sampled lane-entry expectation

- Monte Carlo the released binary gates under steering noise (heading random walk, $\sigma_\omega = \text{CUTIN_W_SD_MODEL}$, the model's own `dt`) along the 12 cut-in clips; compare the sampled conflict-probability with `lane_entry_weight`. Accept: max absolute difference reported per clip; agreement within ~0.15 through the response window validates the closed form, larger structured disagreement goes to review.

Cards E.2, E.3 and D

Deliberately not carded for cheap execution: E.2 (uncertainty term) and E.3 (policy-set field) involve model design, and D (safety audit) touches the causation runner's intervention injection. These get cards after their design notes exist.

4 Suggested session pattern

One card per session, in card order within a stage; start each session with "read `handover.md`, `docs/czb_validation_roadmap.md`, and `docs/czb_work_orders.md`, then execute card X and nothing else"; end each with the card's report, a commit whose message names the card, and one paragraph appended to the running log in `replication/czb/out/worklog.md`. Review sessions (stronger model) after A.2, A.3, and B.4 at minimum.